Solution Requirements

Date	5-7-2026
Team ID	
Project Name	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:
Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	**Requirement Category**	**Requirement Description**	**Priority (High/Medium/Low)**
1	Authentication	*Users can securely access the application to use AI learning features.*	High
2	Authorization levels	*All users can access Question Answering, Quiz, Summary, Explanation, and Learning Recommendation modules.*	High
3	External interfaces	*Integrates Google Gemini 2.5 Flash API to generate AI-powered educational content.*	High

4	Transactions processing	*User requests are processed by FastAPI and AI-generated responses are returned in real time.*	High
5	Reporting	*Displays answers, explanations, quizzes, summaries, and learning recommendations on the web interface.*	Medium
6	Business rules, etc.	*Generates responses based on the user's input and selected learning module.*	High
7	Compliance to laws or regulations	*Protects API keys and user information by following basic security and privacy practices.*	Medium
8	*Other*	*Stores user queries, summaries, quizzes, and learning data in JSON files for future use.*	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	*The application should generate AI responses quickly.*	*Response time within 2–5 seconds*
2	Scalability	The system should support future expansion and additional users.	*Supports 100+ concurrent users*
3	Security & Data Privacy	*API keys and stored data should be protected using environment variables and secure storage.*	*Secure .env configuration*
4	Reliability & Availability	*The application should operate with minimal failures during normal usage.*	*99% uptime*
5	Usability & Accessibility	*The interface should be simple, responsive, and easy to use on desktop and mobile devices.*	*Responsive and user-friendly UI*
6	*Other*	*The code should be modular and easy to maintain or extend with new features.*	*Maintainable and reusable code*